Question	Answer	Marks	Partial Marks
3(d)(i)	$y = 4 - x$ oe final answer	**2**	**B1** for $4 - x$ or $y = k - x$ or $y = 4 + kx$ oe
3(d)(ii)	correct ruled line	**1**	**FT** if in form $y = mx + c$ oe ($m, c \neq 0$)
	0.125 to 0.2 and 2.15 to 2.2	**2**	**B1** for each
4(a)	$[\pm]\sqrt{k - s}$ final answer	**2**	**M1** for $t^2 = k - s$
4(b)(i)	$(x - 5)(x + 5)$ final answer	**1**	
4(b)(ii)	$\frac{x-5}{x-7}$ nfww final answer	**3**	**M2** for $(x - 7)(x + 5)$ or **M1** for $x(x + 5) - 7(x + 5)$ or $x(x - 7) + 5(x - 7)$ or $(x + a)(x + b)$ where $a + b = -2$ or $ab = -35$
4(c)	$\frac{4x^2 - 7x - 8}{x(x+1)}$ or $\frac{4x^2 - 7x - 8}{x^2 + x}$ final answer	**3**	**M1** for $(x - 8)(x + 1) + 3x \times x$ oe isw **B1** for common denominator $x(x + 1)$ oe isw
4(d)	3, 4, 5, 6 nfww	**3**	**B2** for 3 correct or 4 correct and 1 extra or **M2** for $n > \frac{18}{8}$ oe and $n \leqslant 6$ or **M1** for $18 < 8n$ [$\leqslant 30 + 3n$] or [$18 - 3n <$] $5n \leqslant 30$ seen
5(a)(i)	1930 or 1940 or 1933.4 to 1935.3	**5**	**B1** for interior angle 120 soi or angle at centre 60 soi or for correct use of Pythagoras' with 7 and 3.5 or with 14 and 7 **M3** for $6 \times \frac{1}{2} \times 7^2 \times \sin 60 \times 15.2$ oe or complete other methods or **M2** for $6 \times \frac{1}{2} \times 7^2 \times \sin 60$ oe OR **M1** for $\frac{1}{2} \times 7^2 \times \sin 60$ oe or other partial area of hexagon **M1dep** for *their* area × 15.2 evaluated

Question	Answer	Marks	Partial Marks
5(a)(ii)	893 or 892.8 to 893.0…	**3**	**M2** for $6 \times 7 \times 15.2 + 2 \times 6 \times \frac{1}{2} \times 7^2 \times \sin 60$ oe or for 6 × 7 × 15.2 + 2 × *their* area of hexagon from **(a)** oe or **M1** for [6 ×] 7 × 15.2 oe or 2 × *their* area of hexagon from **(a)** oe
5(b)	2.71 or 2.709 to 2.710	**3**	**M2** for $\sqrt[3]{500 \div \left(6 \times \frac{4}{3}\pi\right)}$ oe or **M1** for $500 = 6 \times \frac{4}{3}\pi r^3$ oe If 0 scored, **SC1** for answer 4.92 or 4.923 to 4.924
6(a)	$y > x$	**1**	
	$x \geqslant 15$	**1**	
	$y < 50$	**1**	
	$x + y \leqslant 70$	**1**	
6(b)	Four correct ruled lines and correct region indicated	**5**	all lines ruled **B1** for $y = x$ broken **B1** for $x = 15$ **B1** for $y = 50$ broken **B1** for $x + y = 70$
6(c)	189	**2**	**M1** for (21, 49) seen or for $2x + 3y$ written for a point (x, y) in *their* region where x and y are integers
7(a)(i)	$\frac{9}{160}$ oe	**1**	
7(a)(ii)	58.125 nfww	**4**	**M1** for mid-points soi **M1** for use of Σfx with x in correct interval including both boundaries **M1** (dep on 2nd M1) for $\Sigma fx \div 160$
7(b)	[3 42] 85 140 151 160	**2**	**B1** for 1 error **FT** other values